

AFRILANG Stdlib

std/c/netchecksum

Français. Bibliothèque AFRILANG 'std/c/netchecksum' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : ipChecksum, onesComplement, fold32, verify, pseudoHeader, tcpChecksum.

```
'import "std/c/netchecksum"' · 'use netchecksum'
```

- 'ipChecksum(data list number) → number' - 'onesComplement(n number) → number' - 'fold32(sum number) → number' - 'verify(data list number, expected number) → bool' - 'pseudoHeader(src list number, dst list number, proto number, len number) → list number' - 'tcpChecksum(header list number, payload list number) → number'

English.

AFRILANG library 'std/c/netchecksum' (tier complex, category complex). Exports 6 function(s): ipChecksum, onesComplement, fold32, verify, pseudoHeader, tcpChecksum.

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'import "std/c/netchecksum"' · 'use netchecksum'
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Catégorie / Category	Modules complex (c/)	
Niveau / Tier	complex	June 16, 2026
Fonctions / Functions	6	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/netchecksum' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : ipChecksum, onesComplement, fold32, verify, pseudoHeader, tcpChecksum.

```
'import "std/c/netchecksum"' · 'use netchecksum'
```

```
- `ipChecksum(data list number) → number`
- `onesComplement(n number) → number`
- `fold32(sum number) → number`
- `verify(data list number, expected number) → bool`
- `pseudoHeader(src list number, dst list number, proto number, len number) → list
  number`
- `tcpChecksum(header list number, payload list number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/netchecksum"
use netchecksum
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- ipChecksum(data list number) → number•
onesComplement(n number) → number•
fold32(sum number) → number•
verify(data list number, expected number) → bool•
pseudoHeader(src list number, dst list number, proto number, len number) → list•
tcpChecksum(header list number, payload list number) → number

4. Exemples d'utilisation

```
import "std/c/netchecksum"
use netchecksum
```

```
create result = ipChecksum(/* args */)
say result
```

```
create result = onesComplement(/* args */)
say result
```

```
create result = fold32(/* args */)
say result
```

```
create result = verify(/* args */)
say result
```

```
create result = pseudoHeader(/* args */)
say result
```

```
create result = tcpChecksum(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/netchecksum"`` en tête de fichier.
- Lancez ``afrilang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module netchecksum

```
function ipChecksum(data list number) returns number
end
```

```
function onesComplement(n number) returns number
end
```

```
function fold32(sum number) returns number
end
```

```
function verify(data list number, expected number) returns bool
end
```

```
function pseudoHeader(src list number, dst list number, proto number, len number)
  returns list number
end
```

```
function tcpChecksum(header list number, payload list number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/netchecksum' (tier complex, category complex). Exports 6 function(s): ipChecksum, onesComplement, fold32, verify, pseudoHeader, tcpChecksum.

'import "std/c/netchecksum"' · 'use netchecksum'

```
- `ipChecksum(data list number) → number`
- `onesComplement(n number) → number`
- `fold32(sum number) → number`
- `verify(data list number, expected number) → bool`
- `pseudoHeader(src list number, dst list number, proto number, len number) → list
  number`
- `tcpChecksum(header list number, payload list number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/netchecksum"
use netchecksum
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- ipChecksum(data list number) → number•
- onesComplement(n number) → number•
- fold32(sum number) → number•
- verify(data list number, expected number) → bool•
- pseudoHeader(src list number, dst list number, proto number, len number) → list•
- tcpChecksum(header list number, payload list number) → number

4. Usage examples

```
import "std/c/netchecksum"
use netchecksum
```

```
create result = ipChecksum(/* args */)
say result
```

```
create result = onesComplement(/* args */)
say result
```

```
create result = fold32(/* args */)
say result
```

```
create result = verify(/* args */)
say result
```

```
create result = pseudoHeader(/* args */)
say result
```

```
create result = tcpChecksum(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/netchecksum"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module netchecksum

```
function ipChecksum(data list number) returns number
end
```

```
function onesComplement(n number) returns number
end
```

```
function fold32(sum number) returns number
end
```

```
function verify(data list number, expected number) returns bool
end
```

```
function pseudoHeader(src list number, dst list number, proto number, len number)
  returns list number
end
```

```
function tcpChecksum(header list number, payload list number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/netchecksum"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/netchecksum/>

Source (extrait)

```
module netchecksum

function ipChecksum(data list number) returns number
end

function onesComplement(n number) returns number
end
```

```
function fold32(sum number) returns number
end

function verify(data list number, expected number) returns bool
end

function pseudoHeader(src list number, dst list number, proto number, len number)
  returns list number
end

function tcpChecksum(header list number, payload list number) returns number
end
end
```